

# 1 PROBLEM

A polynomial  $P(x)$  is called "good" if it has integer coefficients and:

- it is monic of degree 7;
- the absolute values of all of its coefficients are odd and strictly less than  $2025^2$ ;
- the absolute values of all of its coefficients are roots of  $P$ . How many good polynomials are there?

# 2 ANSWER

6 150 968.

Equivalently, this number is  $(3 * 2025^2 + 61) / 2$ .

# 3 ===== COMPLETE PROOF

## 3.1 Notation.

Write  $P(x) = c_7 x^7 + c_6 x^6 + c_5 x^5 + c_4 x^4 + c_3 x^3 + c_2 x^2 + c_1 x + c_0$ , with all  $c_i$  integers and  $c_7 = 1$  (monic of degree 7).

Since the absolute value of every coefficient is required to be ODD, every coefficient is a nonzero odd integer. In particular  $c_i \neq 0$  for all  $i$ .

Let  $M := \max_{\{0 \leq i \leq 7\}} |c_i|$ . Because every  $|c_i|$  is odd,  $M$  is a positive odd integer, so  $M = 1$  or  $M \geq 3$ .

Two coefficient-absolute-values are roots automatically by the conditions:

(R1)  $|c_7| = 1$  is a coefficient absolute value, hence 1 is a root:  $P(1) = 0$ .

(R2)  $M = |c_j|$  for some index  $j$ , and that absolute value must be a root, hence  $P(M) = 0$ .

We will also repeatedly use the parity fact: every  $c_i$  and  $M$  are odd, so any integer combination of them whose value we compute can be reduced mod 2; e.g.  $M + c_i$  is even,  $c_i * M + c_k$  is even, etc.

The whole count splits into the two cases  $M = 1$  and  $M \geq 3$ .

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## 3.2 CASE 1: $M = 1$

Here every  $|c_i| = 1$ , so every coefficient is +1 or -1, with  $c_7 = 1$ . The only absolute value occurring among the coefficients is 1, so the ONLY root that the third condition forces is the value 1; that is, the conditions are satisfied exactly when  $P(1) = 0$ .

Now  $P(1) = c_7 + c_6 + \dots + c_0$  is a sum of eight numbers each equal to  $\pm 1$ . This sum is 0 iff exactly four of the eight coefficients equal +1 and four equal -1. Since  $c_7 = +1$  is fixed, we must choose which three of the remaining seven coefficients  $c_0, \dots, c_6$  are equal to +1 (the other four being -1).

Number of choices =  $C(7,3) = 35$ .

Each such choice gives a distinct good polynomial, and every  $M = 1$  good polynomial arises this way. Hence Case 1 contributes exactly

### 3.3 CASE 2: $M \geq 3$ ( $M$ odd)

Throughout this case  $|c_i| \leq M$  for all  $i$ , and  $P(M) = 0$  by (R2).

We first show that the coefficients are completely determined, up to three sign choices, by  $M$ .

Lemma 1.  $c_6 = -M$ .

Proof. From  $P(M) = 0$ ,  $M^7 + c_6 M^6 = -(c_5 M^5 + c_4 M^4 + \dots + c_1 M + c_0)$ , i.e.  $M^6(M + c_6) = -(c_5 M^5 + \dots + c_0)$ . Taking absolute values and using  $|c_i| \leq M$ ,  $M^6 |M + c_6| \leq \sum_{i=0}^5 |c_i| M^i \leq M \sum_{i=0}^5 M^i = \sum_{j=1}^6 M^j$ . Therefore  $|M + c_6| \leq \sum_{j=1}^6 M^j / M^6 = 1 + 1/M + 1/M^2 + \dots + 1/M^5$ . For  $M \geq 3$  the tail satisfies  $1/M + \dots + 1/M^5 \leq 1/3 + 1/9 + 1/27 + 1/81 + 1/243 < 1/2$ , so  $|M + c_6| < 3/2$ . But  $M$  and  $c_6$  are both odd, so  $M + c_6$  is an even integer; an even integer of absolute value  $< 3/2$  is 0. Hence  $c_6 = -M$ .  $\square$

Because  $c_6 = -M$ , the two top terms cancel in  $P(M)$ :  $M^7 + c_6 M^6 = 0$ . Thus  $P(M) = 0$  reduces to

$$(E5) \quad c_5 M^5 + c_4 M^4 + c_3 M^3 + c_2 M^2 + c_1 M + c_0 = 0.$$

Lemma 2.  $c_5 = \pm 1$  and  $c_4 = -c_5 M$ .

Proof. Group (E5) as  $M^4(c_5 M + c_4) = -(c_3 M^3 + c_2 M^2 + c_1 M + c_0)$ . Taking absolute values with  $|c_i| \leq M$ ,  $M^4 |c_5 M + c_4| \leq M(M^3 + M^2 + M + 1)$ , so  $|c_5 M + c_4| \leq (M^3 + M^2 + M + 1)/M^3 = 1 + 1/M + 1/M^2 + 1/M^3$ . For  $M \geq 3$  this is at most  $1 + 1/3 + 1/9 + 1/27 = 40/27 < 3/2$ . Since  $c_5 M + c_4$  is even (both factors odd), it must be 0:  $c_4 = -c_5 M$ . Now  $|c_4| \leq M$  forces  $|c_5| * M \leq M$ , i.e.  $|c_5| \leq 1$ ; as  $c_5$  is a nonzero odd integer,  $c_5 = \pm 1$ , and then  $|c_4| = M$ .  $\square$

Substituting  $c_4 = -c_5 M$  back into (E5):  $c_5 M^5 + c_4 M^4 = c_5 M^5 - c_5 M^5 = 0$ , so (E5) collapses to

$$(E3) \quad c_3 M^3 + c_2 M^2 + c_1 M + c_0 = 0.$$

Lemma 3.  $c_3 = \pm 1$  and  $c_2 = -c_3 M$ .

Proof. Group (E3) as  $M^2(c_3 M + c_2) = -(c_1 M + c_0)$ . Then  $M^2 |c_3 M + c_2| \leq M(M + 1)$ , so  $|c_3 M + c_2| \leq (M+1)/M = 1 + 1/M$ . For  $M \geq 3$  this is  $\leq 4/3 < 3/2$ , and  $c_3 M + c_2$  is even, hence 0:  $c_2 = -c_3 M$ . Then  $|c_2| \leq M$  forces  $|c_3| \leq 1$ , so  $c_3 = \pm 1$  and  $|c_2| = M$ .  $\square$

Substituting into (E3):  $c_3 M^3 + c_2 M^2 = 0$ , so (E3) collapses to

$$(E1) \quad c_1 M + c_0 = 0, \quad \text{i.e.} \quad c_0 = -c_1 M.$$

Since  $|c_0| \leq M$ , we get  $|c_1| \leq 1$ , so  $c_1 = \pm 1$  and  $|c_0| = M$ .

Summary of the forced shape (write  $s = c_5$ ,  $t = c_3$ ,  $u = c_1$ , each  $\pm 1$ ):

$$\begin{aligned} c_7 &= 1, & c_6 &= -M, \\ c_5 &= s, & c_4 &= -sM, \\ c_3 &= t, & c_2 &= -tM, \\ c_1 &= u, & c_0 &= -uM. \end{aligned}$$

Thus  $P(x) = x^7 - Mx^6 + sx^5 - sMx^4 + tx^3 - tMx^2 + ux - uM = x^6(x - M) + sx^4(x - M) + tx^2(x - M) + u(x - M) = (x - M)(x^6 + sx^4 + tx^2 + u)$ . (\*)

So necessarily  $P$  has the form  $(*)$  with  $M \geq 3$  odd and  $s, t, u$  in  $\{+1, -1\}$ . Note the multiset of coefficient absolute values is exactly  $\{1, M\}$ .

Now we impose the remaining requirements and count.

- $M < 2025^2$  is required since  $M = |c_6|$  must be strictly less than  $2025^2$ .
- $M$  is a root of  $(*)$  automatically (factor  $x - M$ ); good, since  $M = |c_6|$  etc. must be a root.
- 1 must be a root by (R1). From  $(*)$ ,  $P(1) = (1 - M)(1 + s + t + u)$ . Since  $M \geq 3$ , the factor  $(1 - M) \neq 0$ , so  $P(1) = 0$  forces  $s + t + u = -1$ . For  $s, t, u$  in  $\{\pm 1\}$ , the sum  $s + t + u$  lies in  $\{-3, -1, 1, 3\}$ ; it equals  $-1$  exactly when exactly one of  $s, t, u$  is  $+1$  and the other two are  $-1$ . There are  $C(3,1) = 3$  such sign patterns.

Conversely, take ANY odd  $M$  with  $3 \leq M < 2025^2$  and any of these 3 sign patterns. Then the polynomial  $(*)$  has integer coefficients, is monic of degree 7, every coefficient absolute value lies in  $\{1, M\}$  (both odd, both  $< 2025^2$ ), and both 1 and  $M$  are roots (1 because  $s+t+u = -1$ ,  $M$  from the factor  $x-M$ ). Hence it is good. So these are precisely the good polynomials with  $M \geq 3$ .

Distinct  $(M, s, t, u)$  give distinct polynomials ( $M$  is read off as  $-c_6$ , and the signs as  $c_5, c_3, c_1$ ), so there is no double counting.

Count for Case 2. The admissible values of  $M$  are the odd integers with  $3 \leq M < 2025^2$ . Since  $2025^2 = 4\,100\,625$  is odd, the odd integers in  $[1, 2025^2)$  are  $1, 3, 5, \dots, 4\,100\,623$ , of which there are  $(2025^2 - 1)/2 = 2\,050\,312$ ; removing  $M = 1$  leaves  $(2025^2 - 3)/2 = 2\,050\,311$  admissible values of  $M$ . Each contributes 3 polynomials, so Case 2 contributes  $3 * 2\,050\,311 = 6\,150\,933$  good polynomials.

### 3.4 TOTAL

The cases  $M = 1$  and  $M \geq 3$  are exhaustive and disjoint (they are distinguished by  $\max|c_i|$ ). Adding the two counts:

$$35 + 6\,150\,933 = 6\,150\,968 .$$

In closed form,  $35 + 3 * (2025^2 - 3)/2 = (3 * 2025^2 + 61)/2 = 6\,150\,968 .$

## 4

### CONCLUSION

The number of good polynomials is

$$6\,150\,968 .$$

A worked example of a good polynomial ( $M = 3, (s,t,u) = (1,-1,-1)$ ):  $P(x) = (x - 3)(x^6 + x^4 - x^2 - 1) = x^7 - 3x^6 + x^5 - 3x^4 - x^3 + 3x^2 - x + 3$ , with coefficient absolute values  $\{1, 3, 1, 3, 1, 3, 1, 3\}$ ; both 1 and 3 are roots ( $P(1) = P(3) = 0$ ), all coefficients are odd, and all are  $< 2025^2$ .